(c) H.M. for Continuous data:

The Harmonic mean H.M. = $\dfrac{N}{\sum_{i=l}^{n} f_i\left(\dfrac{1}{x_i}\right)}$

Where x_i is the mid-point of the class interval

Example 5.13

Find the harmonic mean of the following distribution of data

Dividend yield (percent)	2 – 6	6 – 10	10 – 14
No. of companies	10	12	18

Solution:

Class Intervals	**Mid-value (x_i)**	**No. of companies (f_i)**	**Reciprocal ($1/x_i$)**	$f_i\ (1/x_i)$
2 – 6	4	10	¼	2.5
6 – 10	8	12	1/8	1.5
10 – 14	12	18	1/12	1.5
Total		N = 40		5.5

The harmonic mean is H.M. = $\dfrac{N}{\sum_{i=l}^{n} f_i\left(\dfrac{1}{x_i}\right)} = \dfrac{40}{5.5} = 7.27$

Merits of H.M:

- It is rigidly defined
- It is based on all the observations of the series
- It is suitable in case of series having wide dispersion
- It is suitable for further mathematical treatment
- It gives less weight to large items and more weight to small items

Limitations of H.M:

- It is difficult to calculate and is not understandable
- All the values must be available for computation
- It is not popular due to its complex calculation.
- It is usually a value which does not exist in series

Ch5_11th Stats.indd 125 2/3/2019 11:00:56 AM